

Problem

Calculate the complex Fourier series for $f(t) = e^t$, $-\pi < t < \pi$, with $f(t + 2\pi n) = f(t)$ for all integer values of n

Step-by-step solution

Step 1 of 2

$$\begin{aligned}
 T &= 2\pi, \quad \omega_o = \frac{2\pi}{T} = 1 \\
 f(t) &= \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_o t} \\
 C_n &= \frac{1}{T} \int_0^T f(t) e^{-jn\omega_o t} dt \\
 &= \frac{1}{2\pi} \int_0^{2\pi} e^t e^{-jn\pi t} dt \\
 &= \frac{1}{2\pi} \left[e^{(1-jn)\pi} dt \right]_{-\pi}^{\pi} = \frac{1}{2\pi} \left[\frac{e^{(1-jn)\pi}}{(1-jn)} \right]_{-\pi}^{\pi} \\
 &= \frac{1}{2\pi} \left[\frac{e^{(1-jn)\pi}}{(1-jn)} - \frac{e^{-(1-jn)\pi}}{(1-jn)} \right] \\
 &= \frac{1}{2\pi} \left[\frac{\sin(1-jn)\pi}{(1-jn)} \right] 2j \\
 &= \frac{j \sin(1-jn)\pi}{\pi(1-jn)}
 \end{aligned}$$

Step 2 of 2

$$\begin{aligned}
 C_n &= \frac{1}{2\pi} \left[\frac{e^{\pi} e^{-jn\pi} - e^{-\pi} e^{-jn\pi}}{(1-jn)} \right] \\
 &= \frac{1}{2\pi} \left[\frac{e^{\pi} \cos n\pi - e^{-\pi} \cos n\pi}{(1-jn)} \right] \\
 &= \frac{1}{2\pi} [e^{\pi} - e^{-\pi}] \frac{\cos n\pi}{(1-jn)} \\
 &= \frac{\sinh \pi \cos n\pi}{\pi(1-jn)} \\
 f(t) &= \sum_{n=-\infty}^{\infty} \frac{\sinh \pi \cos n\pi}{\pi(1-jn)} e^{jnt} \\
 \boxed{f(t) = \frac{\sinh \pi}{\pi} \sum_{n=-\infty}^{\infty} \frac{\cos n\pi}{(1-jn)} e^{jnt}}
 \end{aligned}$$